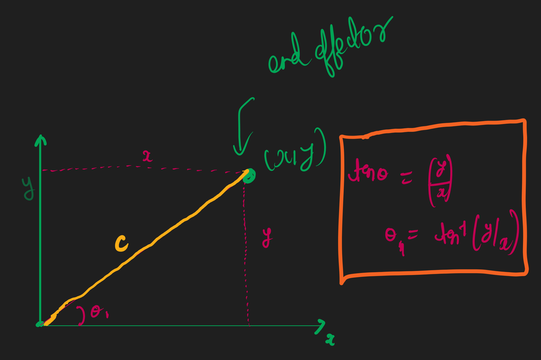
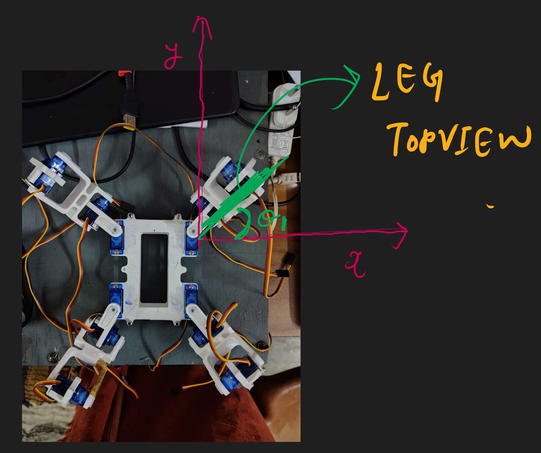
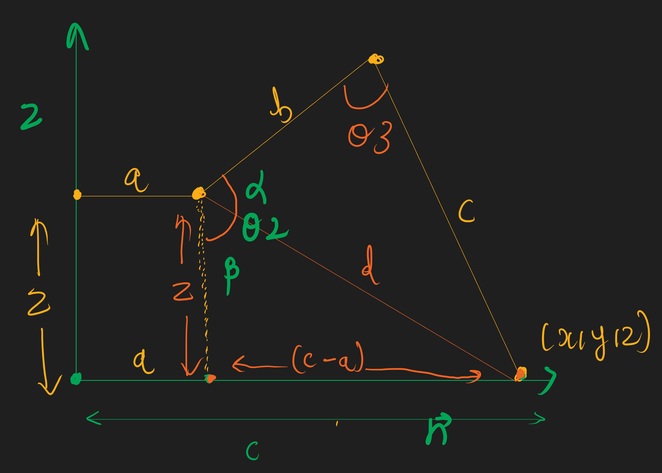


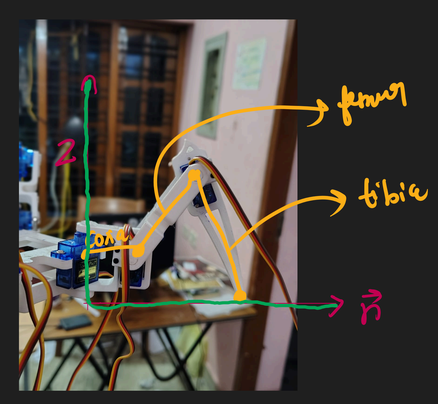
$$\tan\theta = \left(\frac{y}{x}\right)$$

$$\theta_1 = \tan^{-1}(y/x)$$

$$c = \sqrt{x^2+y^2}$$







Coxa, femur, tibia

↳ These are known distance which are depend on the cad file / ACTUAL length.

coxa has length = a mm

femur has length = b mm

tibia has length = c mm.

$$d = \sqrt{(c-a)^2 + z^2}$$

$$\boxed{\theta_2 = \alpha + \beta}$$

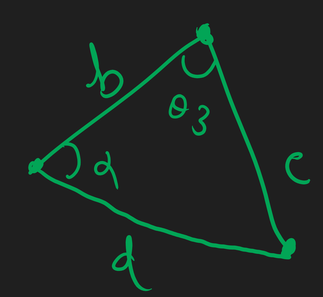
$$\boxed{\theta_3 =}$$

Find α

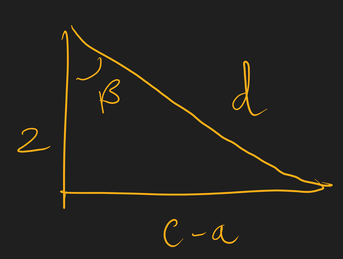
Find α

Hint:

By using cosine rule →



Hint



→ u can use any inverse $\tan^{-1}$, $\cos^{-1}$, $\sin^{-1}$

[only for β
α → cosine rule]